

Empirical Bayes: Step Count Problem

Statistical Reinforcement Learning Lab

Problem Statement

In our study, we have a population of 30 unique individuals and are interested in investigating the number of steps each person takes in a 30-minute interval. We designate the true mean steps for each of 30 individuals as $\mu_1, \mu_2, \dots, \mu_i$ for $i = 1, 2, \dots, 30$ ($n = 30$).

During the study, we use a device to measure the number of steps each individual takes during a 30-minute interval. We use the device to perform this measurement once for each individual.

Now, we have two data points for each individual:

1. The device's step count for that individual.
2. A measure of uncertainty provided by the device, which describes how sure the device is that it counted the number of steps correctly.

We designate the observed step counts as $x_1, x_2, \dots, x_i$ and the measures of uncertainty as $\sigma_1, \sigma_2, \dots, \sigma_i$. Thus, our data for each of the 30 individuals consists of one pair of values, (x_i, σ_i) .

The true means and hyperparameters are unknown to us. The observed step counts and measures of uncertainty are our data and thus are known to us. How can we estimate μ_0 and τ^2 using this data?

Empirical Bayes Solution

We will approach this problem using Raudenbush & Bryk's empirical Bayes model for meta-analyses [1].

First, we assume that each person's true mean steps is a draw from the same underlying normal distribution with some non-zero variance τ^2 :

$$\mu_i \sim \mathcal{N}(\mu_0, \tau^2).$$

We also assume each person's step count on a given day is a draw from a normal distribution centered at that person's mean, μ_i , with non-zero variance σ_i^2 . (We assume that the device's uncertainty measurement, σ_i^2 , accurately captures the true within-person variance.)

Then, for this problem, τ^2 designates population variance while σ_i^2 designates within-person variance.

By equation 8 (p. 78) and equation 18 (p. 80) [1], we can model each individual's observation is a draw from a normal distribution centered at that person's mean μ_i . The variance of this observation is the sum of within-person and population variance. Therefore, the distribution of the observed data $\mathbf{x}$ is:

$$x_i \sim \mathcal{N}(\mu_0, \sigma_i^2 + \tau^2), \quad x_i \mid \mu_i \sim \mathcal{N}(\mu_i, \sigma_i^2).$$

Let us note that our model parallels Raudenbush & Bryk's model as follows [1]: $d_i = x_i$, $W'_i = 1$, $\gamma = \mu_0$, $v_i = \sigma_i^2$. Recall that in our problem, σ_i^2 is known to us as it is provided by the device, but τ^2 and μ_0 are both unknown. Thus, we are in the scenario with unknown variance and will reference the *Variance estimation* section (pp. 80-82).

We can derive the likelihood and log-likelihood functions of τ^2 and μ_0 from the normal PDF as follows:

$$L(\mu_0, \tau^2) = \prod_{i=1}^n \mathcal{N}(x_i | \mu_0, \sigma_i^2 + \tau^2) \quad (1)$$

$$L(\mu_0, \tau^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi(\sigma_i^2 + \tau^2)}} \exp\left[-\frac{(x_i - \mu_0)^2}{2(\sigma_i^2 + \tau^2)}\right] \quad (2)$$

$$\ell(\mu_0, \tau^2) = -\frac{1}{2} \sum_{i=1}^n \left\{ \log(\sigma_i^2 + \tau^2) + \frac{(x_i - \mu_0)^2}{\sigma_i^2 + \tau^2} \right\} \quad (3)$$

We can rewrite the conditional log-likelihood as follows:

$$Q(\mu_0; \tau^2) = \sum_{i=1}^n w_i(\tau^2) (x_i - \mu_0)^2, \quad w_i(\tau^2) = \frac{1}{\sigma_i^2 + \tau^2} \quad (4)$$

Next, we differentiate w.r.t. μ_0 to determine the MLE estimator for μ_0 conditional on τ^2 . Here, we are considering τ^2 as a constant and hence the $\log(\sigma_i^2 + \tau^2)$ term is dropped.

$$\frac{\partial Q}{\partial \mu_0} = -2 \sum_{i=1}^n w_i(\tau^2) (x_i - \mu_0) = 0 \quad (5)$$

$$\mu_0^*(\tau^2) = \frac{\sum_{i=1}^n w_i(\tau^2) x_i}{\sum_{i=1}^n w_i(\tau^2)} \quad (6)$$

We have effectively written the conditional MLE of μ_0 in terms of τ^2 , so we only have one unknown variable remaining (τ^2). We will replace μ_0 with our conditional MLE in the original log-likelihood equation. Now, we have:

$$\ell_p(\tau^2) = -\frac{1}{2} \left\{ \sum_{i=1}^n \log(\sigma_i^2 + \tau^2) + Q_1(\tau^2) \right\}, \quad Q_1(\tau^2) = \sum_{i=1}^n w_i(\tau^2) (x_i - \mu_0^*(\tau^2))^2 \quad (7)$$

We define the MLE of τ^2 as:

$$\hat{\tau}^2 = \arg \max_{\tau^2 \geq 0} \ell_p(\tau^2) \quad (8)$$

Now, we can substitute this estimate back into our conditional MLE estimator for μ_0 to obtain an estimate for μ_0 :

$$\hat{\mu}_0 = \mu_0^*(\hat{\tau}^2) = \frac{\sum_{i=1}^n (\sigma_i^2 + \hat{\tau}^2)^{-1} x_i}{\sum_{i=1}^n (\sigma_i^2 + \hat{\tau}^2)^{-1}} \quad (9)$$

Now, we can write the posterior likelihood function for μ_i as follows:

$$L(\mu_i; \mathbf{x}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left\{-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right\} \cdot \frac{1}{\sqrt{2\pi\tau^2}} \exp\left\{-\frac{(\mu_i - \mu_0)^2}{2\tau^2}\right\} \quad (10)$$

$$\ell(\mu_i; \mathbf{x}) = -\frac{1}{2} \sum_{i=1}^n \left[\log(2\pi\sigma_i^2) + \frac{(x_i - \mu_i)^2}{\sigma_i^2} + \log(2\pi\tau^2) + \frac{(\mu_i - \mu_0)^2}{\tau^2} \right] \quad (11)$$

First, we take the partial derivative w.r.t. μ_i to derive the posterior mean, substituting in the MLE estimators we derived above.

$$\frac{\partial \ell(\mu_i; \mathbf{x})}{\partial \mu_i} = \frac{x_i - \mu_i}{\sigma_i^2} + \frac{\mu_0 - \mu_i}{\tau^2} \quad (12)$$

$$\Rightarrow \mu_i \left(\frac{1}{\sigma_i^2} + \frac{1}{\hat{\tau}^2} \right) = \frac{x_i}{\sigma_i^2} + \frac{\hat{\mu}_0}{\hat{\tau}^2} \quad (13)$$

$$\Rightarrow \mu_i = \lambda_i x_i + (1 - \lambda_i) \hat{\mu}_0, \quad \lambda_i = \frac{\hat{\tau}^2}{\sigma_i^2 + \hat{\tau}^2} \quad (14)$$

Second, we derive the MLE of the posterior variance by Fisher Information:

$$\frac{\partial^2 \ell}{\partial \mu_i^2} = - \left(\frac{1}{\sigma_i^2} + \frac{1}{\tau^2} \right) \quad (15)$$

$$\Rightarrow \text{Var}(\mu_i | x_i) = \left(-\frac{\partial^2 \ell}{\partial \mu_i^2} \right)^{-1} = \left(\frac{1}{\sigma_i^2} + \frac{1}{\tau^2} \right)^{-1} \quad (16)$$

To conclude, we have:

$$\text{Prior: } \mu_i \sim \mathcal{N}(\mu_0, \tau^2) \quad (17)$$

$$\text{Posterior: } \mu_i | \mathbf{x} \sim \mathcal{N} \left(\lambda_i x_i + (1 - \lambda_i) \hat{\mu}_0, \left(\frac{1}{\sigma_i^2} + \frac{1}{\hat{\tau}^2} \right)^{-1} \right), \quad \lambda_i = \frac{\sigma_i^2}{\sigma_i^2 + \hat{\tau}^2}. \quad (18)$$

References

- [1] Stephen W. Raudenbush and Anthony S. Bryk. Empirical bayes meta-analysis. *Journal of Educational Statistics*, 10(2):75–98, 1985.